



Name: _____

Total Time: 30 Minutes

Marks _____ / **35**

Show working and answers on this sheet. Show working in sufficient detail to support your answers. Incorrect answers without supporting reasoning may be allocated zero marks.

Resources Allowed – 1 A4 Page (1 Side) of Notes,

Q1) Simplify the following, leaving in Index form.

a $2^7 \times 2^3$

2^{10} ✓

b $7^4 \div 7^3$

7 ✓

3

c $(2^3)^4$

2^{12} ✓

Q2) Simplify the following

a $k^3 \times k^4$

k^7 ✓

b $\frac{m^{11}}{m^6}$

m^5 ✓

4

c $2y \times y^4 \times y^5$

$2y^{10}$ ✓

d $(x^4)^2$

x^8 ✓

Q3) Evaluate the following problems

a 2^{-3}

$= \frac{1}{8}$ ✓

b 7^{-1}

$= \frac{1}{7}$ ✓

c $\frac{1}{5^{-2}}$

$= 25$ ✓

d $(3)^0$

$= 1$ ✓

5

e $5y^0$

$= 5$ ✓

12

Q4) Simplify the following

a $a^2b^4 \times a^3b$

a^5b^5

b $4n^2 \times 3n \times n^5$

$12n^8$

c $\frac{p^6q^8}{p^3q}$

p^3q^7

6

Q5) Simplify each of the following

a $6ab^3 \times 5a^2b^6$

$30a^3b^9$

b $\left(\frac{x^3}{y^2}\right)^0$

$= 1$ ✓

c $(n^3)^2 \times (2n^3)^3$

$n^6 \times 8n^9$

$8n^{15}$ ✓

d $4s^2 \times 6(s^2t)^3$

$4s^2 \times 6s^6t^3$

$24s^8t^3$ ✓

9

Q6) Leave the answers with positive indices

a $\frac{54x^4y^3}{9x^6y^2} \div 9$

$\frac{6x^4y^3}{x^6y^2} =$

$\frac{6y}{x^2}$

3

b $\frac{2(a^2b)^3}{8(ab)^4} \times \frac{3a^7}{a^9}$

$\frac{2a^6b^3}{8a^4b^4} \times \frac{3a^7}{a^9}$

$= \frac{a^2}{4b} \times \frac{3}{a^2} = \frac{3a^2}{4a^2b}$

$= \frac{3}{4b}$

✓✓✓ working

✓✓ Answer

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